

STA304H1 F: Surveys, Sampling and Observational Data

First Sub-Session, 2026 | University of Toronto

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Lectures: Tuesday 2:00 PM – 5:00 PM ES 1050
Thursday 2:00 PM – 5:00 PM PB B150
Office hours: Fridays 9:00 AM – 10:00 AM Online

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Course Description

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Design of surveys, sources of bias, randomized response surveys. Techniques of sampling: stratification, clustering, unequal probability selection. Sampling inference, estimates of population mean and variances, ratio estimation. Observational data: correlation vs. causation, missing data, sources of bias.

Learning objectives

By the end of this course, students will be able to:

1. Design surveys and sampling strategies to appropriately gather information of interest, including selecting and implementing suitable sampling techniques and study designs, while evaluating trade-offs among alternative approaches.
2. Apply a variety of statistical methods to draw valid inferences from data collected through surveys or samples.
3. Evaluate sources of bias in a study and critically assess study design, including its strengths, limitations, and appropriateness of analytical methods.
4. Effectively communicate statistical results and conclusions to both technical and non-technical audiences.

Quercus Information

This Course uses the University's learning management system, Quercus, to post information about the course. This includes posting readings and other materials required to complete class activities and course assignments, as well as sharing important announcements and updates. New information and resources will be posted regularly as we move through the term. To access the course website, go to the U of T Quercus log-in page at <https://q.utoronto.ca>.

SPECIAL NOTE ABOUT GRADES POSTED ONLINE: Please also note that any grades posted are for your information only, so you can view and track your progress through the course. No grades are considered official, including any posted in Quercus at any point in the term, until they have been formally approved and posted on ACORN at the end of the course. Please contact me as soon as possible if you think there is an error in any grade posted on Quercus.

Class Format

We will meet **in person** on Tuesdays from 2:00 PM to 5:00 PM and Thursdays from 2:00 PM to 5:00 PM. Lectures will be delivered using slide decks, and annotated slides will be posted on Quercus after each class. On both term test days, a lecture video (approximately one hour in length) will be posted on Quercus for students to watch prior to the following Thursday's in-person lecture.

Note: The instructor will make every effort to provide virtual access to in-person lectures. When possible, lectures will be recorded and made available to all students in the course.

The instructor office hours will be on zoom on Fridays from 9:00 AM - 10:00 AM. Invitation links for office hours will be posted on Quercus. Please ensure you are able to access your UofT Zoom account as you will need to be signed in this way to enter any STA304/1003 course related online Zoom meetings <https://utoronto.zoom.us/>.

Course administrative email

Use sta304@course.utoronto.ca for all administrative and personal inquiries. Please write from your @mail.utoronto.ca address to avoid spam filters.

Note: This inbox will not be monitored after July 1, 2026.

After that date, contact emily.somerset@mail.utoronto.ca.

Textbooks and Reference materials

All course materials are provided on Quercus. The supplementary textbooks below are:

1. Sharon L. Lohr *Sampling: Design and Analysis*. Chapman & Hall/CRC, 2021.
2. Richard L. Scheaffer, William Mendenhall, III, R. Lyman Ott, Kenneth G. Gerow *Elementary Survey Sampling, 7th Edition*. Boston: Pws-Kent, 2011.
3. Nathan Taback *Design and Analysis of Experiments and Observational Studies using R*. Chapman & Hall/CRC, 2022.

Assessments and Grading

Grading Scheme

The assessments for STA304 are listed in table below. Your final grade will be calculated as the maximum of Weight Scheme A and Weight Scheme B.

Assessment	Weight scheme A	Weight scheme B	Due Date / Details
Learning activities	Best 8/10 10%	Best 8/10 10%	Generally due Tues/Thurs at 11:59 PM ET (see calendar).
Term Test 1	25%	15%	Thursday, May 21, 2:30 PM – 4:30 PM ET (during class time) • Location TBA on Quercus.
Term Test 2	15%	25%	Thursday, June 4, 2:30 PM – 4:30 PM ET (during class time) • Location TBA on Quercus.
Final Exam	50%	50%	To be scheduled by the Faculty of Arts & Science. • Do not book travel until the official schedule is released. • Conflicts/deferrals: artsci.utoronto.ca.
Total	100%	100%	

Learning activities

On most lecture days, a learning activity will be posted on Quercus (see the course calendar). These low-stakes assessments generally cover the topic(s) of that day's lecture and are designed to help you engage more deeply with the material. It is encouraged to complete these independently, without the use of LLMs—these activities are designed to give you a space to practice and make mistakes while receiving meaningful feedback from the teaching team.

Activities are graded out of 2 points, tentatively as follows:

0 — Missing or irrelevant

- No submission, or response is clearly unrelated to the activity

1 — Minimal engagement

- Incomplete (e.g., only part of the activity is attempted)
- Responses are extremely vague
- Limited evidence of engagement with the material

2 — Genuine attempt (full marks)

- Completes all or most of the activity
- Responses are connected to the course material (even if incorrect)
- Demonstrates effort to apply relevant concepts

Only your best 8 out of 10 activities will count toward your final grade.

Term Tests and Final Exam

All three exams will be held in person. Term Test 1 is 2 hours long and will take place from 2:30 PM to 4:30 PM on Thursday, May 21, 2026, during regular class time. Term Test 2 is 2 hours long and will take place from 2:30 PM to 4:30 PM on Thursday, June 4, 2026, during regular class time. Any updates to the schedule will be announced on Quercus in advance of the test date. The final exam will be 3-hours long and will be scheduled by the Faculty of Arts and Science during the final assessment period in June.

Re-marking Policy

Mistakes occasionally happen when marking. If you feel there is an issue with the marking of an assessment, you may request that it be re-marked. The course re-mark policy exists to correct mistakes, and any request should clearly identify the error (for example, a question that was not marked, or a total that was incorrectly calculated). Re-mark requests must be made via a form (the link will be shared in a Quercus announcement when the grades are posted). Re-mark requests will NOT be accepted by email and must satisfy the following requirements:

- must not be sent within the first 24 hours of the release of the assessment grade,
- must be received **within one week** of the date that the marks for the assessment became available,
- must give specific, clear, and consist reason for each request, referring to a possible error or omission by the marker. Re-mark requests without a specific reason will not be considered.

Please note that your entire test/Assignment may be remarked when submitting a re-mark request. Keep in mind that it is possible for your assessment grade to go down if the regraded mark is lower than your original assessment grade. All re-marking will be done by the instructor.

Note: Final exam regrading follows Faculty of Arts & Sciences procedures.

Course Calendar

Course schedule

Note: The schedule is subject to change. All deadlines are at 11:59 PM ET unless otherwise stated.

Date	Lec.	Topics	Deliverables
May 5	L1	• Elements of the sampling problem • Selection bias • Measurement error • Questionnaire design • Ethics in research	LA1 posted
May 7	L2	• Framework for probability sampling • Types of probability samples • Simple random sample (SRS)	LA2 posted
May 12	L3	• Inference under SRS for population mean, total, and proportion. • When SRS should be used	LA3 posted LA1 due LA2 due
May 14	L4	• Inference under stratified sampling (STRS) for population mean, total, and proportion. • Allocating observations to strata	LA4 posted LA3 due
May 19	L5	• Defining Strata • Ratio estimation • Regression estimation	LA5 posted LA4 due
May 21	L6	• Poststratification	Term test 1
May 26	L7	• Cluster Sampling	LA6 posted LA5 due
May 28	L8	• Sampling with Unequal probability	LA7 posted LA6 due
June 2	L9	• Nonresponse • Imputation Methods	LA8 posted LA7 due
June 4	L10	• Survey Quality • Randomized Response	Term test 2
June 9	L11	• Observational Data • Correlation vs. causation • Causal Inference	LA9 posted LA8 due
June 11	L12	• Causal effects and observational studies • Propensity scores and methods	LA10 posted LA9 due LA10 due June 14
Final Exam Period			TBA — June

Key: • Learning activity due • Posted **Bold red** Term test

Study halls

Study halls are optional sessions held before each term test. Students are welcome to attend and work either collaboratively or independently. Member(s) of the teaching team will be present to provide support. Changes to these sessions and locations for all sessions will be announced on Quercus.

Before Test 1 Wednesday May 20, 10:00 AM - 4:00 PM & Thursday May 21, 9:00 AM - 12:00 PM

Before Test 2 TBA & Thursday June 4, 9:00 AM - 12:00 PM

Course Policies

Missed Term Work

4.1.1 Policy for learning activities

No additional accommodations are available for missed activities beyond the built-in flexibility (best 8 of 10). Due to the accelerated nature of the course, learning activities submitted after the deadline will not be accepted.

4.1.2 Policy for term tests

This course runs over six weeks, which makes scheduling make-up tests logistically infeasible. To compensate, the course includes two term tests rather than a single midterm, providing multiple opportunities to demonstrate your learning. The term test component (40% total) will be calculated using the most favorable of the following weightings:

- Term Test 1 (25%), Term Test 2 (15%)
- Term Test 1 (15%), Term Test 2 (25%)

If you are unable to write one of the term tests and provide valid documentation, the adjusted weighting will be as follows:

- **Missed Term Test 1 only:** Term Test 2: 35%, Final Exam: 55%, Learning Activities: 10%
- **Missed Term Test 2 only:** Term Test 1: 35%, Final Exam: 55%, Learning Activities: 10%

In the rare case that both term tests are missed with valid documentation, the full weight will be shifted to the final exam (Final Exam: 90%, Learning Activities: 10%). Students in this situation should contact the instructor and their College Registrar as early as possible to discuss academic planning.

If any reweighting results in a final exam worth more than 50% of your final grade, you are encouraged to consult your College Registrar for academic planning support.

To request reweighting for a missed term test, you must submit one of the following recognized forms of documentation:

- Absence Declaration via ACORN (once-per-term limit),
- University of Toronto Verification of Illness or Injury (VOI) form,
- College Registrar's letter,
- Letter of Academic Accommodation from Accessibility Services.

Any one of these on its own is sufficient — additional documentation will not be requested. Detailed medical information beyond the VOI, and letters from friends or family, will not be accepted. For more information on the VOI, please see <http://www.illnessverification.utoronto.ca>. For information on Absence Declaration Tool for A&S students, please see <https://www.artsci.utoronto.ca/absence>. If you get a concussion, break your hand, or suffer some other acute injury, you should register with Accessibility Services as soon as possible.

You must also notify the instructor as soon as possible by email, and no later than one week after the missed test (or one week after returning to class, if your absence is extended). If you do not come forward within this window, reweighting may not be granted except where an accessibility accommodation, religious accommodation, or other documented extended absence applies.

4.1.3 Policy for final exam

The final exam will be scheduled by the Faculty of Arts and Sciences sometime in the final assessment period in June. The final exam will be cumulative and will be a proctored in-person timed examination. Any scheduling conflicts or petitions to write a deferred exam must go through the Faculty of Arts and Sciences standard procedures <https://www.artsci.utoronto.ca/current/faculty-registrar/exams-assessments>. Be sure not to make travel plans until the final exam schedule has been officially released by the Faculty of Arts and Sciences

Academic integrity

All suspected cases of academic dishonesty will be investigated following procedures outlined in the Code of Behaviour on Academic Matters (<https://governingcouncil.utoronto.ca/secretariat/policies/code-behaviour-academic-matters-july-1-2019>). If you have questions or concerns about what constitutes appropriate academic behaviour or appropriate research and citation methods, please reach out to me. Note that you are expected to seek out additional information on academic integrity from me or from other institutional resources. For example, to learn more about how to cite and use source material appropriately and for other writing support, see the U of T writing support website at <http://www.writing.utoronto.ca>. Consult the Code of Behaviour on Academic Matters for a complete outline of the University's policy and expectations. For more information, please see A&S Student Academic Integrity (<https://www.artsci.utoronto.ca/current/academic-advising-and-support/student-academic-integrity>) and the University of Toronto Website on Academic Integrity (<https://www.academicintegrity.utoronto.ca>).

Discussion of lecture material and course concepts with classmates and the teaching team is encouraged, but it is expected that you work independently on all individual assessments. Please note, you may not submit for credit any work that was completed by another student (or person). This includes, but is not limited to, partially or fully completed code, communication of solutions, and plagiarism. In particular, you are expected to complete and submit independent work for learning activities, the term tests, and the final exam. You may use code provided by your STA304/1003 instructors without providing a citation. If you use code from any other source, you must provide the source. To protect yourself from potential academic integrity offences, do not share your code and written submissions.

Students with Disabilities or Accommodation Requirements

Students with diverse learning styles and needs are welcome in this course. If you have an acute or ongoing disability issue or accommodation need, you should register with Accessibility Services (AS) at the beginning of the academic year by visiting <https://studentlife.utoronto.ca>.

[ca/departments/accessibility-services/](https://utoronto.ca/departments/accessibility-services/). Without registration, you will not be able to verify your situation with your instructors, and instructors will not be advised about your accommodation needs. AS will assess your situation, develop an accommodation plan with you, and support you in requesting accommodation for your course work. Remember that the process of accommodation is private: AS will not share details of your needs or condition with any instructor, and your instructors will not reveal that you are registered with AS.

Religious Accommodations

As a student at the University of Toronto, you are part of a diverse community that welcomes and includes students and faculty from a wide range of cultural and religious traditions. For my part, I will make every reasonable effort to avoid scheduling tests, examinations, or other compulsory activities on religious holy days not captured by statutory holidays. Further to University Policy, if you anticipate being absent from class or missing a major course activity (such as a test or in-class assignment) due to a religious observance, please let me know as early in the course as possible, and with sufficient notice (at least two to three weeks), so that we can work together to make alternate arrangements.

Computing

Computational work is a central part of developing statistical thinking and developing facility in the use of computational tools for carrying out simulations and data analysis.

We will use **R**, **RStudio**, and **Quarto**. All of these are freely available. You need to first install R, and then R Studio. R can be downloaded for free from <http://cran.r-project.org>. R Studio can be downloaded for free from <http://www.rstudio.com/products/rstudio/download/>. Additionally, you can also use R Studio through the U of T Jupyterhub, by selecting the RStudio option and logging in with your utorID and password, available here: <https://jupyter.utoronto.ca>.

Some resources for using R and Quarto:

- The course supplementary notes give guidelines on installing and getting started with R and R Studio.
- A short intro to R workshop is available here: <https://awstringer1.github.io/ssu-r-workshop/ssu-r-workshop.html>.
- Hands-On Programming with R by Garrett Grolmund, available here: <https://rstudio-education.github.io/hopr>.
- A Quarto Cheat Sheet is available at <https://rstudio.github.io/cheatsheets/quarto.pdf>.

Communication

4.6.1 Piazza

We will be using Piazza as the platform for discussions related to the course material and assessments. Students can post anonymously to classmates on Piazza, but the identity of the author of all posts is viewable by instructors.

Be sure to read Piazza's Privacy Policy and Terms of Service carefully. Take time to understand and be comfortable with what they say. They provide for substantial sharing and disclosure of your personal information held by Piazza, which affects your privacy. When you use Piazza, only

provide content that you are comfortable sharing under the terms of the Privacy Policy and Terms of Use. With that being said Piazza will still be considered a part of our class and thus all posts and conduct on Piazza must remain professional. Posts regarding personal matters such as inquiries about grades, reporting absences, regrade requests, etc. should be communicated via email (at sta304@course.utoronto.ca) and *NOT* be posted on Piazza. Piazza is intended for students to receive support regarding course information and content and thus should be an overall positive and professional environment. Again, email is appropriate only for personal matters that can not be shared with the rest of the class. To be fair to all students, we are not able to answer questions about the course material by email. These questions should be asked on the discussion forum (publicly) or during office hours.

Inquires about administrative matters, such as missed tests and re-mark requests, should be sent to sta304@course.utoronto.ca. Please note that email will not be monitored on evenings, holidays or weekends (Toronto time). Additionally, Piazza will not be monitored by the teaching team on evenings, holidays or weekends (Toronto time). Thus, the teaching team would like to encourage students to respond to each other's questions on Piazza to allow for quicker response times. Depending on the amount of emails please allow a reasonable amount of time for email (and Piazza) responses. Announcements and other course information will be posted on Quercus.

Course materials

Course materials are provided for the exclusive use of enrolled students. These materials should not be reposted, shared, put in the public domain, or otherwise distributed without the explicit permission of the instructor. These materials belong to your instructor, the University, and/or other sources depending on the specific facts of each situation and are protected by copyright. Students violating these policies will be subject to disciplinary actions under the Code of Student Conduct.

Generative AI

Students may use artificial intelligence tools, including generative AI, as learning aids. However, students are ultimately accountable for the work they submit.

4.8.1 Invigilated Tests

Students may not use generative artificial intelligence tools (e.g. ChatGPT and other AI writing and coding assistants) for the completion of, or to support the completion of invigilated, closed book assessments, including the term tests and final exam.

4.8.2 Learning activities

You are encouraged not to use AI for this component of the course. I don't expect your responses to be perfect. Consider these activities as low-stakes opportunities to exercise deeper engagement with course concepts.

Equity, Diversity and Inclusion

The University of Toronto is committed to equity, human rights and respect for diversity. All members of the learning environment in this course should strive to create an atmosphere of mutual respect where all members of our community can express themselves, engage with each other, and respect one another's differences. U of T does not condone discrimination or harassment against any persons or communities.

Mental health resources

Your mental health is important. Throughout university life, there are many experiences that can impact your mental health and well-being. As a University of Toronto student, you can access free mental health and wellness services at Health & Wellness (<https://studentlife.utoronto.ca/department/health-wellness/>) such as same day counseling, medical care, skill-building workshops, and drop-in peer support. You can also meet with a Wellness Navigation Adviser who can connect you with other campus and community services and support. Call the mental health clinic at 416-978-8030 ext. 5 to book an appointment or visit <https://uoft.me/mentalhealthcare> to learn about the services available to you.

You can also visit your College Registrar to learn about the resources and supports available: <https://www.artsci.utoronto.ca/current/academic-advising-and-support/college-registrars-offices>

If you're in distress, you can access immediate support: <https://uoft.me/feelingdistressed>